

1. Original McKibben force equation

Your equation is the Chou-Hannaforf geometric model.

$$F = \frac{\pi D_0^2 P}{4 \sin^2 \theta_0} [3(1 - \varepsilon)^2 \cos^2 \theta_0 - 1]$$

An equivalent form is:

$$F = \frac{\pi D_0^2 P}{4} \left[\frac{3(1 - \varepsilon)^2}{\tan^2 \theta_0} - \frac{1}{\sin^2 \theta_0} \right]$$

Where:

F = theoretical contraction force

D_0 = initial actuator diameter

P = internal gauge pressure

θ_0 = initial braid angle

ε = contraction ratio

$$\varepsilon = \frac{L_0 - L}{L_0}$$

This is the standard Chou-Hannaforf model. ([PubMed Central \(PMC\)](#))

2. Does this equation change for hydraulic McKibben?

For static force calculation, no fundamental change is required.

You can use:

$$F_{\text{ideal}} = \frac{\pi D_0^2 P}{4 \sin^2 \theta_0} [3(1 - \varepsilon)^2 \cos^2 \theta_0 - 1]$$

for both:

Pneumatic McKibben, air as working fluid

Hydraulic McKibben, oil or water as working fluid

The important variable is the internal pressure P .

The equation itself does not contain air density, oil density, water density, viscosity or bulk modulus.

The operating fluid becomes important when you model dynamic behavior, pressure response, energy efficiency and flow losses.

Experimental work has directly compared pneumatic and hydraulic McKibben muscles and found that hydraulic operation can provide higher efficiency and different hysteresis behavior. ([Sage Journals](#))

3. Why this equation does not accurately predict actual force

The ideal equation assumes:

Ideal braid

Inextensible fibres

No braid friction

No bladder resistance
No braid-bladder friction
No wire bending resistance
No significant end effects

Therefore:

$$F_{\text{experimental}} \neq F_{\text{ideal}}$$

Usually, the ideal model overpredicts the experimentally measured contraction force because it neglects energy losses. ([PubMed Central \(PMC\)](#))

4. Correction for bladder thickness

There is a published improved equation that includes the thickness of the bladder and braid.

It is:

$$F_c = \frac{\pi}{4} P D_0^2 (3 \cos^2 \theta - 1) + \pi P t_k \left[D_0 \left(2 \sin \theta - \frac{1}{\sin \theta} \right) - t_k \right]$$

Here:

t_k = effective thickness term, consisting of the bladder thickness plus approximately half of the sleeve thickness.

This is more realistic than the basic Chou-Hannaford equation because it considers the physical thickness of the actuator structure. ([ResearchGate](#))

Reference:

V. G. Kortman et al., “Designing McKibben muscles: a critical review for practical implementation,” *Bioinspiration & Biomimetics*, 2026.

[DOI: 10.1088/1748-3190/ae66c0](#)

5. Correction for friction and other mechanical losses

For your project, this is probably the most useful extended equation.

The contraction force can be written as:

$$F_c = \frac{\pi}{4} P D_0^2 (3 \cos^2 \theta_0 - 1) - \frac{4 m^2 n}{b \sin \theta_0} (M_f + M_d + M_r + M_{tr})$$

where:

m = number of wires in the braid

n = number of turns

b = length of one braided fibre

M_f = resisting moment due to friction between braid wires

M_d = resisting moment due to wire bending

M_r = resisting moment due to bladder bulging

M_{tr} = resisting moment due to friction between the braid and bladder

This equation is much closer to a real McKibben actuator than the basic equation.

([ResearchGate](#))

Conceptually:

$$F_{\text{actual}} = F_{\text{pressure}} - F_{\text{friction}} - F_{\text{wire}} - F_{\text{bladder}} - F_{\text{other}}$$

6. What about bladder hyperelasticity?

This is where you should not simply add a correction factor to the force equation.

For a rubber bladder, you can model its nonlinear behavior using a hyperelastic material model.

For example, the Neo-Hookean model is:

$$W = C_{10}(I_1 - 3)$$

A two-parameter Mooney-Rivlin model is:

$$W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3)$$

where:

W = strain energy density

C_{10}, C_{01} = experimentally determined material constants

I_1, I_2 = first and second strain invariants

These constants should come from material testing, such as tensile testing of your actual bladder material.

You can then use the hyperelastic model in an FE model.

The calculation becomes:

$$P \rightarrow \text{bladder deformation} \rightarrow D, \theta, L \rightarrow \text{braid deformation/contact} \rightarrow F$$

This is more appropriate when you want to account for the actual bladder material rather than treating it as an ideal membrane.

7. Where does bladder thickness enter?

Bladder thickness affects:

Radial expansion

Bladder stiffness

Required pressure

Braid contact

Braid angle

Contraction

Hysteresis

Therefore, you should not simply insert thickness into the basic pressure-force term.

A better approach is:

$$t_b \rightarrow \text{hyperelastic deformation} \rightarrow D(P), L(P), \theta(P) \rightarrow F$$

The published thickness-corrected equation above is useful for analytical calculations.

For high accuracy, use the thickness and hyperelastic properties in an FE model.

8. Does bladder density matter?

Usually, not for your static force equation.

Bladder density primarily affects:

Mass

Inertia

Dynamic response

Natural frequency

Acceleration

It does not directly create a significant pressure-generated axial force.

Therefore:

$$\rho_b \text{ is generally not required in the static force equation}$$

But it should be included in a dynamic FE model.

9. Does hydraulic fluid density matter?

Again, not significantly for static force.

For static conditions:

$$F \approx f(P, D, \theta, \varepsilon)$$

Fluid density becomes important for:

Fluid inertia

Acceleration

Pressure transients

Flow through valves

Flow through pipes

Dynamic response

For hydraulic modelling, you should consider the fluid bulk modulus:

$$K = -V \frac{dP}{dV}$$

Hydraulic oil has a much higher effective bulk modulus than compressed air.

This is one of the fundamental reasons hydraulic McKibben actuators behave more rigidly and can respond faster than pneumatic ones.

10. Does fluid viscosity matter?

Yes, but mainly in the hydraulic system dynamics.

You can have:

$$\Delta P_{\text{loss}} = \Delta P_{\text{pipe}} + \Delta P_{\text{valve}} + \Delta P_{\text{fitting}}$$

These losses affect the pressure actually reaching the actuator.

Therefore, instead of assuming:

$$P_{\text{pump}} = P_{\text{actuator}}$$

you may need:

$$P_{\text{actuator}} = P_{\text{pump}} - \Delta P_{\text{loss}}$$

This is important for a dynamic hydraulic test setup.

11. Pneumatic versus hydraulic McKibben

Parameter	Pneumatic McKibben	Hydraulic McKibben
Working fluid	Air	Oil or water
Fluid compressibility	High	Very low
Bulk modulus	Low	High
Compliance	High	Low
Dynamic stiffness	Lower	Higher
Position control	More difficult	Better
Response	Generally slower	Generally faster
Energy stored in fluid	High	Much lower
Typical pressure capability	Lower	Higher
Force at identical P and geometry	Similar theoretically	Similar theoretically
Fluid density effect	Mainly dynamic	Mainly dynamic
Fluid viscosity	Usually less important	More important
Bladder compatibility	Relatively simple	Very important
Leakage	Air leakage	Hydraulic fluid leakage
Main advantage	Compliance	High force, stiffness and efficiency
Main disadvantage	Compressibility	High-pressure sealing and fluid compatibility

Meller et al. experimentally compared pneumatic and hydraulic McKibben muscles and tested hydraulic-compatible Viton and LDPE bladder materials. ([Sage Journals](#))

[Meller et al., 2014, DOI 10.1177/1045389X14549872](#)

12. What I recommend for your hydraulic McKibben model

I would structure your calculations like this:

Level 1, basic theoretical model

$$F_{\text{ideal}} = \frac{\pi D_0^2 P}{4 \sin^2 \theta_0} [3(1 - \varepsilon)^2 \cos^2 \theta_0 - 1]$$

Level 2, thickness-corrected model

$$F_{\text{thickness}} = \frac{\pi}{4} P D_0^2 (3 \cos^2 \theta - 1) + \pi P t_k \left[D_0 \left(2 \sin \theta - \frac{1}{\sin \theta} \right) - t_k \right]$$

Level 3, loss-corrected model

$$F_{\text{corrected}} = \frac{\pi}{4} P D_0^2 (3 \cos^2 \theta_0 - 1) - \frac{4 m^2 n}{b \sin \theta_0} (M_f + M_d + M_r + M_{tr})$$

Level 4, nonlinear FE model

Include:

$$P, t_b, C_{10}, C_{01}, \rho_b, \mu, K, \text{contact, friction}$$

This is the appropriate route if your objective is to predict your actual hydraulic actuator rather than only calculate its ideal force.

13. Most important references

Chou and Hannaford, foundational McKibben force model. The model is derived using virtual work and the geometric relationship of the braided sleeve. ([PubMed Central \(PMC\)](#))

Kortman et al., 2026, “Designing McKibben muscles: a critical review for practical implementation.” This is the best recent reference for bladder thickness, friction, wire bending, bladder bulging and braid-bladder friction corrections. ([ResearchGate](#))

Meller, Bryant and Garcia, 2014, “Reconsidering the McKibben muscle: Energetics, operating fluid, and bladder material.” This is directly relevant to hydraulic versus pneumatic operation and bladder material selection. ([Sage Journals](#))

For your project, I would use the equations in this order:

$$\text{Chou-Hannaford} \rightarrow \text{thickness correction} \rightarrow \text{friction/loss correction} \rightarrow \text{hyperelastic FE model}$$

That gives you a defensible progression from simple analytical calculation to a realistic hydraulic McKibben model.

Yes. But I need to correct one important point from my previous answer.

The 4th, 5th, and 6th formulas I gave are not three independent published closed-form equations that can simply be combined. They represent three different modelling levels. The literature does provide derivations for them, but the derivation route is different for each.

1. Bladder-thickness correction

The equation was

$$F_c = \frac{\pi}{4} P D_0^2 (3 \cos^2 \theta - 1) + \pi P t_k \left[D_0 \left(2 \sin \theta - \frac{1}{\sin \theta} \right) - t_k \right]$$

This equation does have a derivation.

Kortman et al. 2026 explicitly identify it as the bladder-thickness correction and attribute the improved formulation to earlier work by Chou and Hannaford. ([ResearchGate](#))

Reference:

V. G. Kortman et al., “Designing McKibben muscles: a critical review for practical implementation,” *Bioinspiration & Biomimetics*, 2026.

[DOI: 10.1088/1748-3190/ae66c0](#)

The derivation starts from force equilibrium of the pressurized cylindrical actuator.

The pressure produces an axial force:

$$F_p = PA$$

For an ideal thin-walled actuator, the pressure contribution and braid contribution are combined using the braid geometry.

The braid geometry gives:

$$\tan \theta = \frac{2\pi r n}{L}$$

and therefore:

$$r = \frac{L \tan \theta}{2\pi n}$$

Using the inextensibility of the braid:

$$b^2 = L^2 + (2\pi r n)^2$$

where b is the length of an individual braid fibre.

Therefore:

$$b^2 = L^2 (1 + \tan^2 \theta)$$

and:

$$b = \frac{L}{\cos \theta}$$

For the initial configuration:

$$b = \frac{L_0}{\cos \theta_0}$$

Thus:

$$L \cos \theta = L_0 \cos \theta_0$$

which gives:

$$L = L_0 \frac{\cos \theta_0}{\cos \theta}$$

Since:

$$\varepsilon = \frac{L_0 - L}{L_0}$$

we obtain:

$$1 - \varepsilon = \frac{L}{L_0}$$

and consequently:

$$\cos \theta = \frac{\cos \theta_0}{1 - \varepsilon}$$

This is the geometric foundation of the force equation.

The thickness correction then comes from treating the bladder and braid as having finite radial thickness instead of treating the actuator as an infinitely thin membrane.

The important parameter is:

$$t_k = t_b + \frac{t_s}{2}$$

where:

t_b = bladder thickness

t_s = effective sleeve/braid thickness

The resulting correction is the equation above. ([DOI](#))

2. There is another very useful thickness/material derivation

For your project, I actually recommend looking at Kothera et al. rather than relying only on the compact Kortman equation.

Kothera, Jangid, Sirohi and Wereley developed a static McKibben model that explicitly considers bladder material behavior and thickness.

Their work derives force relationships using:

Pressure force

Rubber stress

Braid tension

Geometric constraints

Bladder deformation

They obtain force expressions from equilibrium equations such as:

$$PRL = \sigma_c tL + NT \cos \alpha$$

and

$$F + \pi R^2 P = \sigma_z A_{\text{rubber}} + NT \sin \alpha$$

These are then combined with the rubber constitutive relation and geometric constraints to obtain the actuator force. ([ResearchGate](#))

This is much closer to what you are asking for because the rubber material enters the derivation explicitly.

Reference:

K. S. Kothera, M. Jangid, J. Sirohi and M. W. Wereley, “Experimental Characterization and Static Modeling of McKibben Actuators,” Journal of Mechanical Design, 2009.

This is one of the papers I would recommend for your literature review.

3. Hyperelastic material model

The 6th formula is different.

I previously wrote:

$$W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3)$$

This is not itself a McKibben force equation.

It is a constitutive equation for the bladder material.

It tells you how the rubber stores strain energy when it deforms.

For an incompressible two-parameter Mooney-Rivlin material:

$$W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3)$$

where:

$$I_1 = \lambda_1^2 + \lambda_2^2 + \lambda_3^2$$

and:

$$I_2 = \lambda_1^2 \lambda_2^2 + \lambda_2^2 \lambda_3^2 + \lambda_3^2 \lambda_1^2$$

For incompressibility:

$$\lambda_1 \lambda_2 \lambda_3 = 1$$

Therefore:

$$\lambda_3 = \frac{1}{\lambda_1 \lambda_2}$$

This gives the rubber stress response.

You then combine that material response with pressure loading and braid constraints.

That produces the actuator force.

A McKibben FEM study explicitly uses a two-parameter Mooney-Rivlin model for the rubber tube and notes that the basic Chou-Hannaford equation neglects rubber nonlinearity, wall thickness and friction. ([Wiley Online Library](#))

4. There is a paper specifically deriving McKibben behavior from rubber properties

This paper is particularly relevant to your question:

“Modeling of McKibben type pneumatic artificial rubber muscle considering rubber material properties”

The authors specifically state that their objective is to calculate McKibben characteristics from actual rubber physical properties rather than fitting arbitrary parameters.

They construct an expansion-plane model, derive the force balance, then model rubber elasticity using a Mooney-Rivlin model. ([J-STAGE](#))

Reference:

“Modeling of McKibben type pneumatic artificial rubber muscle considering rubber physical properties”

Transactions of the JSME, Vol. 88, No. 905, 2022.

\boxed{\text{DOI: 10.1299/transjsme.21-00286}}

[JSME paper](#)

5. What about the friction formula?

This is where I need to make another distinction.

The equation I previously gave was:

$$F_c = \frac{\pi}{4} P D_0^2 (3 \cos^2 \theta_0 - 1) - \frac{4 m^2 n}{b \sin \theta_0} (M_f + M_d + M_r + M_{tr})$$

The four terms represent:

M_f = wire-wire friction

M_d = wire bending resistance

M_r = bladder bulging resistance

M_{tr} = braid-bladder friction

Kortman et al. explicitly discuss these four mechanisms as energy losses. ([ResearchGate](#))

But you should not treat this equation as a universally validated closed-form equation for every McKibben actuator.

It is a loss formulation based on resisting moments.

The basic idea is:

$$\boxed{F_{\text{actual}} = F_{\text{ideal}} - F_{\text{loss}}}$$

with:

$$F_{\text{loss}} = \frac{4 m^2 n}{b \sin \theta} (M_f + M_d + M_r + M_{tr})$$

The resisting moments must themselves be determined from the contact/friction/material mechanics.

So you cannot calculate F_{actual} simply by knowing bladder thickness.

6. The most important new paper for your 6th formula

There is a very recent paper that is particularly relevant:

Gupta, Bose and Santapuri, “Large Deformation Modeling of McKibben Actuators With Helical Inextensibility and End Effects,” Journal of Applied Mechanics, 2026.

The authors formulate a pressurized hyperelastic tube with a helical inextensibility constraint.

Their model includes:

Hyperelastic tube + pressure + helical braid constraint + large deformation + end effects

They use a Mooney-Rivlin hyperelastic model and compare the reduced analytical model with an axisymmetric finite-element model. ([DOI](#))

Reference:

D. Gupta, B. P. Bose and S. Santapuri, “Large Deformation Modeling of McKibben Actuators With Helical Inextensibility and End Effects,” Journal of Applied Mechanics, 93(8), 081002, 2026.

[DOI: 10.1115/1.4071970](#)

7. The derivation route for your actual hydraulic actuator

For your project, I would not try to put every parameter into one giant equation.

Instead, derive it in layers.

Start with the geometric model:

$$F_{\text{ideal}} = \frac{\pi D_0^2 P}{4} \left[\frac{3(1 - \varepsilon)^2}{\tan^2 \theta_0} - \frac{1}{\sin^2 \theta_0} \right]$$

Then include finite thickness:

$$F_{\text{thickness}} = F_{\text{ideal}} + \Delta F_t$$

where:

$$\Delta F_t = \pi P t_k \left[D_0 \left(2 \sin \theta - \frac{1}{\sin \theta} \right) - t_k \right]$$

Then include material elasticity:

$$W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3)$$

Then calculate rubber stresses from W .

Then include braid/contact/friction:

$$F_{\text{actual}} = F_{\text{pressure}} - F_{\text{rubber}} - F_{\text{friction}} - F_{\text{wire}} - F_{\text{end}}$$

This is much more physically meaningful than simply writing:

$$F = f(P, D, \theta, \varepsilon, t, \rho, C_{10}, C_{01}, \mu, \dots)$$

because each term has a physical origin.

8. One important correction for your hydraulic project

The 2026 Gupta et al. paper is primarily a McKibben actuator model for a pressurized hyperelastic tube with a braid constraint. It is not specifically a hydraulic-oil McKibben derivation. ([DOI](#))

However, the mechanical formulation is applicable to the actuator structure itself.

For your hydraulic actuator, replace the pneumatic pressure input with hydraulic pressure:

$$P = P_{\text{hydraulic}}$$

For quasi-static analysis, the fluid density does not need to appear in the force equation.

For dynamic analysis, add:

$$\rho_f$$

fluid viscosity:

$$\mu_f$$

and bulk modulus:

$$K_f$$

to the hydraulic fluid model.

9. Which references I recommend you actually use

For your research, I would prioritize these four:

1. Chou and Hannaford

Use it for the fundamental geometric force derivation.

2. Kothera et al., 2009

Use it for bladder elasticity, thickness and static force modelling. ([ResearchGate](#))

3. JSME 2022, DOI 10.1299/transjsme.21-00286

Use it for a derivation that explicitly incorporates actual rubber material properties through Mooney-Rivlin modelling. ([J-STAGE](#))

4. Gupta, Bose and Santapuri, 2026

Use it for modern hyperelastic, large-deformation and braid-inextensibility modelling. ([DOI](#))

And use Kortman et al. 2026 as the review that connects the geometric model, thickness correction and energy-loss mechanisms. ([DOI](#))

If your objective is to derive a final force equation specifically for your hydraulic McKibben actuator, the next step should be to derive it from first principles using your actual parameters, D_0 , L_0 , θ_0 , bladder thickness, Mooney-Rivlin constants, braid geometry and hydraulic pressure. That derivation can be written step-by-step from pressure work and force equilibrium rather than using an empirical correction factor.

Yes, but there is an important distinction. There is no single universally accepted McKibben force equation that directly contains fluid density, bladder bulk modulus, hyperelastic constants, friction, viscosity, thickness, and braid geometry all at once.

For your hydraulic McKibben actuator, the physically correct approach is a coupled fluid-structure model.

A useful formulation is:

$$F_{\text{out}} = F_{\text{pressure}} - F_{\text{bladder}} - F_{\text{friction}} - F_{\text{braid}} - F_{\text{end}}$$

with the individual terms obtained from the equations below.

1. Pressure-generated force

Start with the standard McKibben equation:

$$F_P = \frac{\pi D_0^2 P}{4 \sin^2 \theta_0} [3(1 - \varepsilon)^2 \cos^2 \theta_0 - 1]$$

This is the main force-producing term.

2. Bladder material

For a hyperelastic bladder, use a strain-energy function.

For Mooney-Rivlin:

$$W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3)$$

where

$$I_1 = \lambda_1^2 + \lambda_2^2 + \lambda_3^2$$

$$I_2 = \lambda_1^2 \lambda_2^2 + \lambda_2^2 \lambda_3^2 + \lambda_3^2 \lambda_1^2$$

and for an incompressible rubber:

$$\lambda_1 \lambda_2 \lambda_3 = 1$$

The bladder stress is then obtained from the strain-energy function.

For example:

$$\sigma_i = \lambda_i \frac{\partial W}{\partial \lambda_i} - p$$

where p is the incompressibility pressure multiplier.

This is where your hyperelastic constants C_{10} and C_{01} enter the model.

3. Bladder compressibility

If you specifically mean the bulk modulus of the bladder material, introduce:

$$K_b = -V_b \frac{dP_b}{dV_b}$$

For a compressible hyperelastic material, the strain-energy function can contain a volumetric term:

$$W = W_{\text{iso}} + W_{\text{vol}}$$

A common form is:

$$W = C_{10}(\bar{I}_1 - 3) + C_{01}(\bar{I}_2 - 3) + \frac{K_b}{2}(J - 1)^2$$

where:

K_b = bladder bulk modulus

J = volume ratio

$$J = \lambda_1 \lambda_2 \lambda_3$$

This allows the bladder to have finite compressibility.

However, for typical rubber-like bladder materials, treating the material as nearly incompressible is often appropriate, so K_b is usually much larger than the shear modulus.

4. Hydraulic-fluid bulk modulus

This is different from bladder bulk modulus.

For hydraulic oil or water:

$$K_f = -V_f \frac{dP}{dV_f}$$

For small pressure changes:

$$\Delta V_f = -\frac{V_f}{K_f} \Delta P$$

This affects the dynamic pressure response.

A simple hydraulic continuity equation is:

$$Q = \frac{dV_{act}}{dt} + \frac{V_f}{K_f} \frac{dP}{dt} + Q_{leak}$$

This is extremely useful for your hydraulic actuator.

Here:

Q = flow entering actuator

V_{act} = actuator internal volume

V_f = fluid volume

K_f = hydraulic fluid bulk modulus

Q_{leak} = leakage flow

5. Fluid density

Fluid density does not normally appear in the static force equation.

It enters the dynamic hydraulic equation through fluid inertia.

For example, a simplified pressure-drop relation can be written as:

$$\Delta P = \frac{\rho_f L}{A} \frac{dQ}{dt} + \Delta P_{friction} + \Delta P_{local}$$

where:

ρ_f = hydraulic fluid density

L = hydraulic line length

A = hydraulic line cross-sectional area

Q = flow rate

Therefore:

$$\rho_f \rightarrow \text{fluid inertia} \rightarrow \text{dynamic pressure} \rightarrow F_{\text{out}}$$

This is why fluid density should not simply be multiplied into your McKibben force equation.

6. Hydraulic viscosity

Pressure losses in the hydraulic circuit can be represented as:

$$\Delta P_{\text{loss}} = \Delta P_{\text{pipe}} + \Delta P_{\text{valve}} + \Delta P_{\text{fitting}}$$

For laminar flow in a cylindrical tube:

$$\Delta P = \frac{8\mu_f L Q}{\pi R^4}$$

where:

μ_f = fluid dynamic viscosity

This gives:

$$P_{\text{actuator}} = P_{\text{pump}} - \Delta P_{\text{loss}}$$

The actuator force should use P_{actuator} , not necessarily pump pressure.

7. Braid friction

A real actuator has friction between braid fibres and between the braid and bladder.

You can represent this as:

$$F_{\text{friction}} = F_{\text{wire-wire}} + F_{\text{wire-bladder}}$$

A commonly used resisting-moment formulation is:

$$F_{\text{friction}} = \frac{4m^2 n}{b \sin \theta} (M_f + M_{tr})$$

where:

m = number of braid wires

n = number of braid turns

b = fibre length

M_f = wire-wire friction moment

M_{tr} = braid-bladder friction moment

Kortman et al. discuss this loss formulation and the associated mechanisms in detail.

8. A practical combined equation

For your hydraulic McKibben actuator, I would formulate the model as:

$$F_{\text{out}} = \frac{\pi D_0^2 P_{\text{act}}}{4 \sin^2 \theta_0} [3(1 - \varepsilon)^2 \cos^2 \theta_0 - 1] - F_{\text{bladder}} - F_{\text{friction}} - F_{\text{braid}} - F_{\text{end}}$$

where:

$$P_{\text{act}} = P_{\text{pump}} - \Delta P_{\text{loss}}$$

and:

$$Q = \frac{dV_{\text{act}}}{dt} + \frac{V_f}{K_f} \frac{dP}{dt} + Q_{\text{leak}}$$

while the bladder behavior is determined from:

$$W = C_{10}(\bar{I}_1 - 3) + C_{01}(\bar{I}_2 - 3) + \frac{K_b}{2}(J - 1)^2$$

and:

$$F_{\text{friction}} = \frac{4m^2n}{b \sin \theta} (M_f + M_{tr} + \dots)$$

This is a coupled model rather than one single algebraic equation.

9. Where each parameter enters

Parameter	Where it enters
P	Force equation
D_0	Force equation
θ_0	Force equation
ε	Force equation
t_b	Bladder model
C_{10}, C_{01}	Hyperelastic bladder model
K_b	Compressible bladder model
K_f	Hydraulic dynamic model
ρ_f	Fluid inertia
μ_f	Flow/pressure losses
Q	Hydraulic continuity
m, n, b	Braid/friction model
μ_{contact}	Braid friction
Leakage	Hydraulic continuity
End effects	Structural model
Bladder density ρ_b	Dynamic structural model

10. The important correction for your project

Do not make an equation such as:

$$F = f(P, D, \theta, \varepsilon, \rho_f, K_f, K_b, C_{10}, C_{01}, t_b, \mu_f, \mu_{\text{contact}})$$

and multiply all these parameters together.

That would not have a physical derivation.

Instead, your model should be:

Hydraulic system $\rightarrow P(t), Q(t)$

$P(t), Q(t) \rightarrow$ bladder deformation

bladder deformation + braid constraint + friction $\rightarrow F(t)$

This is the scientifically defensible approach.

11. References

The most relevant references for this formulation are:

Kothera et al., 2009, “Experimental Characterization and Static Modeling of McKibben Actuators.” This is useful for the static force derivation and incorporation of bladder mechanics. (researchgate.net)

Meller, Bryant and Garcia, 2014, “Reconsidering the McKibben muscle: Energetics, operating fluid, and bladder material.” This directly addresses pneumatic versus hydraulic operation and bladder materials. (journals.sagepub.com)

Kortman et al., 2026, “Designing McKibben muscles: a critical review for practical implementation.” This is particularly useful for thickness, friction, bladder resistance and modelling limitations. (doi.org)

Gupta, Bose and Santapuri, 2026, “Large Deformation Modeling of McKibben Actuators With Helical Inextensibility and End Effects.” This is useful for the hyperelastic, large-deformation and FE formulation. (doi.org)

JSME, 2022, “Modeling of McKibben type pneumatic artificial rubber muscle considering rubber physical properties.” This is particularly useful if you want to connect experimentally obtained rubber properties to the actuator model. (jstage.jst.go.jp)

For your hydraulic actuator, the strongest approach would therefore be to use the Chou-Hannaford equation for the baseline force, Kothera for static bladder mechanics, the JSME/Mooney-Rivlin formulation for rubber properties, and a hydraulic continuity equation containing K_f , ρ_f , μ_f , and leakage for the dynamic model.

You do not normally calculate C_{10} and C_{01} directly from density, thickness, or hardness. They are material constants obtained by fitting experimental stress-strain data for your actual bladder material.

For your McKibben actuator, the most reliable method is a tensile test followed by Mooney-Rivlin fitting.

1. Mooney-Rivlin equation

For a two-parameter incompressible Mooney-Rivlin material:

$$W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3)$$

The constants C_{10} and C_{01} have units of pressure, usually MPa. They are experimentally determined material parameters. ([PubMed Central \(PMC\)](#))

2. What test do you need?

For your bladder material, perform a uniaxial tensile test.

You need:

Initial specimen length L_0

Initial cross-sectional area A_0

Applied force F

Current length L

Then calculate engineering stress:

$$\sigma_{\text{eng}} = \frac{F}{A_0}$$

and stretch ratio:

$$\lambda = \frac{L}{L_0}$$

For rubber, the stretch ratio is much more useful than conventional engineering strain for hyperelastic fitting.

3. Convert the test data

Assuming incompressibility:

$$\lambda_2 = \lambda_3 = \frac{1}{\sqrt{\lambda}}$$

The Mooney-Rivlin uniaxial stress equation becomes:

$$\sigma = 2 \left(\lambda - \frac{1}{\lambda^2} \right) \left(C_{10} + \frac{C_{01}}{\lambda} \right)$$

This equation is directly suitable for fitting your tensile-test data. ([ScienceDirect](#))

4. Rearrange it to find C_{10} and C_{01}

Divide both sides by:

$$2 \left(\lambda - \frac{1}{\lambda^2} \right)$$

giving:

$$\frac{\sigma}{2(\lambda - \lambda^{-2})} = C_{10} + C_{01}\lambda^{-1}$$

Now define:

$$Y = \frac{\sigma}{2(\lambda - \lambda^{-2})}$$

and:

$$X = \frac{1}{\lambda}$$

Then:

$$Y = C_{10} + C_{01}X$$

This is simply a straight-line equation:

$$Y = b + mX$$

Therefore:

$$C_{10} = b$$

and:

$$C_{01} = m$$

This is called the Mooney-Rivlin plot. The intercept gives C_{10} , and the slope gives C_{01} . This procedure is documented in the rubber mechanics literature. ([PubMed Central \(PMC\)](#))

5. Example

Suppose your tensile test gives:

$$L_0 = 25 \text{ mm}$$

and at one point:

$$L = 50 \text{ mm}$$

Then:

$$\lambda = \frac{50}{25} = 2$$

Suppose the measured stress is:

$$\sigma = 1.50 \text{ MPa}$$

Then:

$$Y = \frac{1.50}{2(2 - 1/2^2)}$$

$$Y = \frac{1.50}{3.5}$$

$$Y = 0.4286 \text{ MPa}$$

and:

$$X = \frac{1}{2} = 0.5$$

You repeat this calculation for every experimental data point.

You then plot:

$$Y$$

against:

$$X = \frac{1}{\lambda}$$

and fit:

$$Y = C_{10} + C_{01}X$$

The intercept is C_{10} .

The slope is C_{01} .

Do not calculate C_{10} and C_{01} from only one data point. You need multiple tensile-test points and regression. Abaqus also recommends least-squares fitting of experimental stress-strain data to determine hyperelastic coefficients. ([Abaqus UCLouvain](#))

6. Better method for your McKibben bladder

For your project, I recommend this procedure:

Step 1

Take the exact bladder material you intend to use.

Step 2

Cut standard rubber tensile specimens.

Step 3

Measure:

$$L_0, W_0, t_0$$

Step 4

Perform tensile testing.

Ideally follow:

ISO 37:2017

Rubber, vulcanized or thermoplastic, determination of tensile stress-strain properties.

Step 5

Record:

$$F, L$$

throughout the test.

Step 6

Calculate:

$$\sigma = \frac{F}{A_0}$$

and:

$$\lambda = \frac{L}{L_0}$$

Step 7

Calculate:

$$Y = \frac{\sigma}{2(\lambda - \lambda^{-2})}$$

and:

$$X = \frac{1}{\lambda}$$

Step 8

Fit:

$$Y = C_{10} + C_{01}X$$

Step 9

Use the obtained:

$$C_{10}$$

and:

$$C_{01}$$

in your FE model.

7. Important point for your hydraulic McKibben

Do not take C10 and C01 from a random rubber datasheet if you want accurate simulation.

- The constants depend on:
- Rubber formulation
- Curing/crosslinking
- Temperature
- Strain rate
- Aging
- Loading direction
- Filler content
- Manufacturing process

Therefore, the best values are obtained from the actual bladder material.

8. What about bulk modulus?

C10 and C01 primarily describe the deviatoric or shear response.

For a compressible Mooney-Rivlin model, you also need a volumetric parameter.

Abaqus writes the model as:

$$W = C_{10}(\bar{I}_1 - 3) + C_{01}(\bar{I}_2 - 3) + \frac{1}{D_1}(J - 1)^2$$

The initial shear modulus is:

$$\mu_0 = 2(C_{10} + C_{01})$$

and:

$$K_0 = \frac{2}{D_1}$$

where K_0 is the initial bulk modulus. ([VT Software Documentation](#))

So your material characterization becomes:

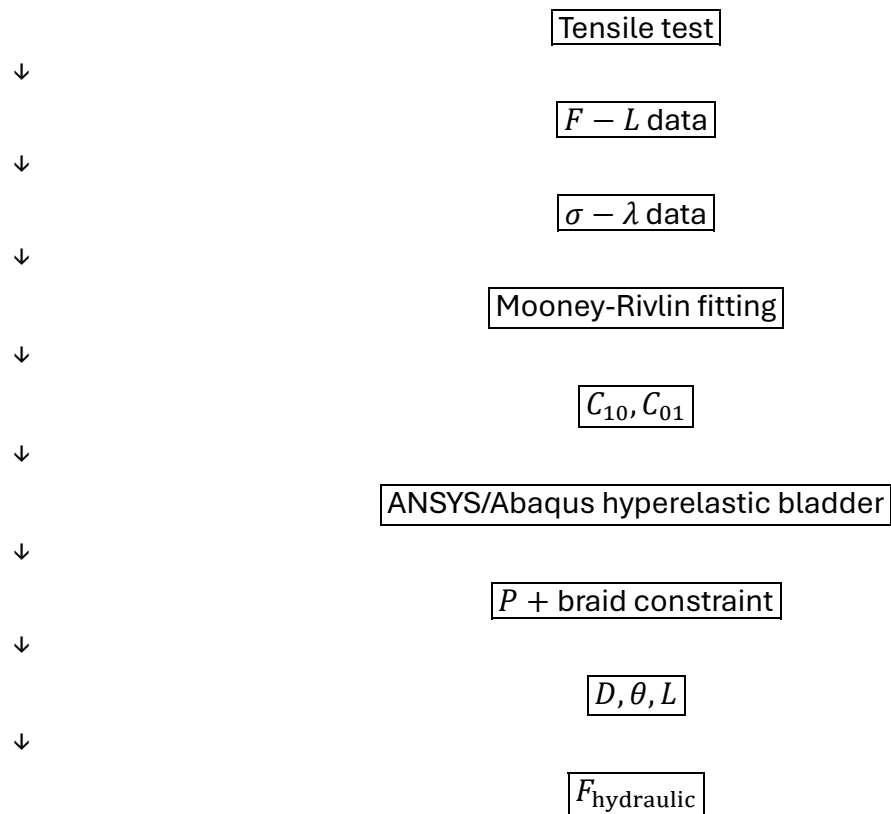
$$\text{Tensile test} \rightarrow C_{10}, C_{01}$$

and, if you need compressibility:

$$\text{Volumetric/compression data} \rightarrow K_b \text{ or } D_1$$

For a rubber bladder, you can often assume near incompressibility, but if you specifically want bulk modulus in your model, you should characterize or otherwise justify it rather than arbitrarily assigning a value.

9. For your project, I would use this workflow



This is much more defensible than assuming C_{10} and C_{01} from a generic rubber property.

References:

Abaqus, Hyperelastic Behavior of Rubberlike Materials, Mooney-Rivlin formulation and parameter fitting. ([VT Software Documentation](#))

Treloar, The Physics of Rubber Elasticity, foundational treatment of rubber stress-strain behavior and Mooney-Rivlin analysis. The Mooney plot method is also documented in the literature. ([PubMed Central \(PMC\)](#))

For your McKibben application, the JSME study on modelling McKibben artificial rubber muscles using actual rubber physical properties is particularly relevant. ([PubMed Central \(PMC\)](#))

If you give me your bladder material, for example NBR, EPDM, silicone, polyurethane, Viton, etc., and its thickness, I can tell you exactly which tensile-test data you need and give you the calculation procedure for obtaining C_{10} , C_{01} , K_b , and the values required in ANSYS.

Below is a consolidated table for the hydraulic McKibben model. I have separated the actual McKibben force equations from the material and hydraulic equations, because they come from different models.

No.	Formula	Purpose	Parameters	Reference
1	$F = \frac{\pi D_0^2 P}{4 \sin^2 \theta_0} [3(1 - \varepsilon)^2 \cos^2 \theta_0 - 1]$	Basic ideal McKibben contraction force	$D_0, P, \theta_0, \varepsilon$	Chou & Hannaford, 1996, “Measurement and Modeling of McKibben Pneumatic Artificial Muscles”
2	$\varepsilon = \frac{L_0 - L}{L_0}$	Contraction ratio	L_0, L	Chou & Hannaford, 1996
3	$b^2 = L^2 + (2\pi r n)^2$	Braid geometry	b, L, r, n	Chou & Hannaford, 1996
4	$\tan \theta = \frac{2\pi r n}{L}$	Braid angle relationship	r, n, L, θ	Chou & Hannaford, 1996
5	$b = \frac{L}{\cos \theta}$	Fibre-length constraint	b, L, θ	Chou & Hannaford, 1996
6	$L \cos \theta = L_0 \cos \theta_0$	Relates current and initial braid geometry	L, L_0, θ, θ_0	Chou & Hannaford, 1996
7	$F_c = \frac{\pi}{4} P D_0^2 (3 \cos^2 \theta - 1) + \pi P t_k \left[D_0 \left(2 \sin \theta - \frac{1}{\sin \theta} \right) - t_k \right]$	Finite bladder/braid thickness correction	t_k, D_0, P, θ	Kortman et al., 2026, review of McKibben modelling

No.	Formula	Purpose	Parameters	Reference
8	$t_k = t_b + \frac{t_s}{2}$	Effective thickness	t_b, t_s	Kortman et al., 2026
9	$W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3)$	Two-parameter Mooney-Rivlin bladder model	C_{10}, C_{01}, I_1, I_2	Mooney, 1940; Rivlin, 1948
10	$I_1 = \lambda_1^2 + \lambda_2^2 + \lambda_3^2$	First strain invariant	$\lambda_1, \lambda_2, \lambda_3$	Mooney-Rivlin theory
11	$I_2 = \lambda_1^2\lambda_2^2 + \lambda_2^2\lambda_3^2 + \lambda_3^2\lambda_1^2$	Second strain invariant	$\lambda_1, \lambda_2, \lambda_3$	Mooney-Rivlin theory
12	$\lambda_1\lambda_2\lambda_3 = 1$	Incompressibility condition for rubber	Principal stretches	Rubber hyperelasticity theory
13	$\sigma = 2\left(\lambda - \frac{1}{\lambda^2}\right)\left(C_{10} + \frac{C_{01}}{\lambda}\right)$	Uniaxial Mooney-Rivlin stress	$\sigma, \lambda, C_{10}, C_{01}$	Mooney-Rivlin uniaxial formulation
14	$Y = \frac{\sigma}{2(\lambda - \lambda^{-2})}$	Mooney plot transformation	σ, λ	Mooney-Rivlin fitting method
15	$Y = C_{10} + C_{01}\lambda^{-1}$	Determine C_{10}, C_{01} by linear regression	$Y, \lambda, C_{10}, C_{01}$	Mooney-Rivlin fitting method
16	$\mu_0 = 2(C_{10} + C_{01})$	Initial shear modulus	C_{10}, C_{01}	Abaqus hyperelastic formulation
17	$W = C_{10}(\bar{I}_1 - 3) + C_{01}(\bar{I}_2 - 3) + \frac{1}{D_1}(J - 1)^2$	Compressible Mooney-Rivlin material	C_{10}, C_{01}, D_1, J	Abaqus, Hyperelastic Material Models
18	$K_0 = \frac{2}{D_1}$	Initial bulk modulus of bladder	K_0, D_1	Abaqus hyperelastic formulation

No.	Formula	Purpose	Parameters	Reference
19	$K_b = -V_b \frac{dP_b}{dV_b}$	Definition of bladder bulk modulus	K_b, V_b, P_b	Continuum mechanics
20	$K_f = -V_f \frac{dP}{dV_f}$	Hydraulic-fluid bulk modulus	K_f, V_f, P	Hydraulic fluid mechanics
21	$\Delta V_f = -\frac{V_f}{K_f} \Delta P$	Fluid compression relationship	$V_f, K_f, \Delta P$	Hydraulic fluid mechanics
22	$Q = \frac{dV_{act}}{dt} + \frac{V_f}{K_f} \frac{dP}{dt} + Q_{leak}$	Hydraulic continuity equation	$Q, V_{act}, V_f, K_f, P, Q_{leak}$	Hydraulic system modelling
23	$\Delta P = \frac{8\mu_f L Q}{\pi R^4}$	Laminar pipe pressure loss	μ_f, L, Q, R	Hagen-Poiseuille equation
24	$P_{act} = P_{pump} - \Delta P_{loss}$	Pressure reaching actuator	$P_{pump}, P_{act}, \Delta P$	Hydraulic circuit modelling
25	$\Delta P_{inertia} = \frac{\rho_f L}{A} \frac{dQ}{dt}$	Fluid inertia contribution	ρ_f, L, A, Q	1-D hydraulic fluid dynamics
26	$F_{friction} = \frac{4m^2 n}{b \sin \theta} (M_f + M_{tr})$	Braid friction resistance	$m, n, b, \theta, M_f, M_{tr}$	Kortman et al., 2026
27	$F_{loss} = \frac{4m^2 n}{b \sin \theta} (M_f + M_d + M_r + M_{tr})$	Combined mechanical loss	M_f, M_d, M_r, M_{tr}	Kortman et al., 2026
28	$F_{actual} = F_{ideal} - F_{loss}$	Practical force estimate	F_{ideal}, F_{loss}	Loss-based McKibben modelling
29	$W = W_{iso} + W_{vol}$	Separate deviatoric and volumetric bladder response	W_{iso}, W_{vol}	Hyperelasticity theory

No.	Formula	Purpose	Parameters	Reference
30	$W = C_{10}(\bar{I}_1 - 3) + C_{01}(\bar{I}_2 - 3) + \frac{K_b}{2}(J - 1)^2$	Hyperelastic bladder including finite compressibility	C_{10}, C_{01}, K_b, J	Compressible Mooney-Rivlin formulation

Important: Formula 30 is a generic compressible Mooney-Rivlin form. If you use ANSYS or Abaqus, use the exact strain-energy convention and parameter definitions of that software rather than mixing K_b and D_1 definitions.

Key references

Reference	What you should use it for
Chou, C. P., and Hannaford, B., 1996, “Measurement and Modeling of McKibben Pneumatic Artificial Muscles,” IEEE Transactions on Robotics and Automation, 12(1), 90–102.	Fundamental McKibben geometry and force derivation
Kothera, C. S., Jangid, M., Sirohi, J., and Wereley, N. M., 2009, “Experimental Characterization and Static Modeling of McKibben Actuators,” Journal of Mechanical Design.	Static modelling, bladder mechanics and experimental characterization
Meller, M. A., Bryant, M., and Garcia, E., 2014, “Reconsidering the McKibben Muscle: Energetics, Operating Fluid, and Bladder Material.”	Hydraulic versus pneumatic operation and bladder materials
Kortman et al., 2026, “Designing McKibben muscles: a critical review for practical implementation,” Bioinspiration & Biomimetics.	Thickness, friction, wire bending, bladder resistance and modelling approaches
Gupta, D., Bose, B. P., and Santapuri, S., 2026, “Large Deformation Modeling of McKibben Actuators With Helical Inextensibility and End Effects,” Journal of Applied Mechanics.	Hyperelastic tube, large deformation, braid constraint and FE modelling
JSME, 2022, “Modeling of McKibben type pneumatic artificial rubber muscle considering rubber physical properties.”	Determining actuator behavior using actual rubber properties
Mooney, M., 1940, “A Theory of Large Elastic Deformation,” Journal of Applied Physics.	Mooney material formulation

Reference	What you should use it for
Rivlin, R. S., 1948, "Large Elastic Deformations of Isotropic Materials IV," Philosophical Transactions of the Royal Society A.	Rivlin hyperelastic formulation
ISO 37:2017	Tensile testing of rubber specimens to obtain stress-strain data
Abaqus Documentation, Hyperelastic Material Models	Practical fitting of C_{10} , C_{01} , bulk modulus and hyperelastic parameters

For your hydraulic McKibben project, the model can therefore be organized as:

Pump -> P, Q, ρ_f , μ_f , K_f -> Actuator Pressure -> Bladder Deformation -> C_{10} , C_{01} , K_b , t_b -> Braid Deformation -> Friction/ Contact -> F_{out}

The most important equations for your actual calculation are Nos. 1, 7, 9, 13, 15, 20, 22, 23, 25, 27, and 30.

One caution: density ρ_f , hydraulic-fluid bulk modulus K_f , and viscosity μ_f belong primarily to the hydraulic dynamic model. They should not be artificially inserted into the static Chou-Hannaford force equation.